

Ayush Yadav

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github.com/blackphoenix42 | linkedin.com/in/ayushyadav | codechef.com/users/blackphoenix42

Skills

Languages: C++, Python, TypeScript, Go, Bash, SQL

AI/ML: LLMs, RAG, Vector Embeddings, AI Agents, MCP, Graph Neural Networks, Deep Learning

Backend: Distributed Systems, REST, gRPC, GraphQL, Docker, CI/CD, Kafka, Pub/Sub, Linux

Cloud & Data: GCP, AWS, PostgreSQL, MongoDB, Redis

EDA & Performance: SystemVerilog, Undo Debugger, Profiling, AddressSanitizer, Valgrind, Perforce, Git

Frontend: React, Redux, Tailwind, Canvas, Accessibility

Experience (~4 Years)

Cadence Design Systems, Noida

July 2022 - Present

R&D Software Engineer II

- Built XMAI, an AI-assisted EDA profiler using **LLMs, RAG, embeddings, MCP**, and **Cadence JEDAI** to improve failure analysis and optimization workflows.
- Optimized **Xcelium Logic Simulator** using low-level C++, scalable profiling, and runtime analysis, improving simulation throughput by **18–19%** on large customer workloads.
- Implemented RTL transformations in the Xform Engine, improving runtime performance by **13–14%** for designs from **Apple, Google, Samsung, and NVIDIA**.
- Built and presented a C++ performance-analysis library at **Cadence India Conference**, enabling Top-N profiling and structured diagnostics that reduced debug **RCA time by ~40%**.
- Earned **Corporate VP / MD recognition** for driving performance wins that led to a **multi-million-dollar Samsung deal**.

Projects

- **XMAI - Agentic AI Profiler**
 - Designed the profiler pipeline to parse simulation artifacts, **identify hotspots**, and generate **RTL optimization recommendations**.
 - Built **CLI, TUI, GUI, MCP Server**, and auto-analyze-on-failure workflows to enable faster debugging and tool integration for large SoC simulation workloads.
 - Developed **RAG pipelines** with vector embeddings, retrieval, tool-calling, prompt orchestration, and graph-compatible RTL representations for EDA triage.
 - Contributed to **ChipStack AI** by building C++ APIs, JEDAI embedding-upload flows, and **Agentic Building Blocks** for composing chip-design agents.
- **AlgoLens - Interactive DSA Platform**
 - Built an **algorithm visualizer** (60+ algos) with real-time step-through, deterministic replay, & **FFmpeg-WASM export**.
 - Added GenAI explanations, **complexity analysis**, and **exact, fuzzy, phonetic, and semantic search**.
 - Improved quality with **WCAG 2.1 AA** accessibility, **Sentry**, Web Vitals, Playwright, Storybook, Percy, and CI gates.

Achievements

- CodeChef **6-star, Rating 2353, AIR 29** - *BlackPhoenix42*.
- Codeforces **Master, Rating 2264, AIR 13** - *BinaryPhoenix10*.
- LeetCode **Knight, Rating 2124, Global Top 1%** - *BinaryPhoenix*.
- Hackerrank **6-star Problem Solver** - *BinaryPhoenix*
- Awarded **\$10,000** to expand a decentralized gaming app on Tezos Blockchain.

Education

Netaji Subhas University of Technology (NSUT) East Campus

Aug 2018 - May 2022

Bachelor of Technology (B.Tech) in Computer Science and Engineering

CGPA: 8.4 / 10